

Print Settings — Bambu Lab X2D

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These mirror **Bambu Studio's built-in presets** — the point is to **pick the right filament profile and plate type**, not hand-type numbers. Use this to sanity-check a profile and know which knob to turn.

PER-FILAMENT SETTINGS · 0.4 MM NOZZLE · TEXTURED PEI PLATE

SETTING	PLA BASIC	PLA MATTE	PETG	ABS
Nozzle temp	220 °C	220 °C	255 °C	270 °C
Bed temp	55 °C	55 °C	70 °C 60–65 on the smooth/Cool plate	100 °C textured PEI only — not the Cool plate
Part-cooling fan	100%	100%	~50% more fan = weaker layers	0–20% off for the first layers; fan = cracks
Max volumetric speed the real speed cap	~21 mm³/s	~12 mm³/s clogs easier	~10 mm³/s	~22 mm³/s high, but chamber heat is the real limit
Enclosure (door + top)	Open	Open	Open closed traps heat → clogs	Closed both shut — let the chamber reach 40–50 °C
Dry the spool?	Rarely	If opened a while	Yes — 65 °C / 8 h + desiccant	If opened a while 80 °C / 8 h
Bed adhesion	Bare plate	Bare plate	Glue stick as <i>release barrier</i> , or smooth plate	Bare hot plate + a brim glue stick if a corner still lifts
Shrinkage part ends up smaller	Negligible	Negligible	Slight	~0.5–0.8% a 100 mm part lands near 99.4 mm
Support top Z-gap	Default	Default	+0.2 mm PETG supports fuse on	Default
Retraction	Leave at preset (~0.8 mm @ 30 mm/s) — fix stringing with drying & temp first.			

COMMON PROCESS SETTINGS — ALL FOUR FILAMENTS

Layer height	0.20 mm standard · 0.28 mm draft · 0.12 mm fine · first layer 0.20 mm
Walls / shells	2 wall loops (3 for functional parts) · 4 top + 4 bottom layers (~0.8 mm)
Infill	15% general · 10% display-only · 25–40% functional · Gyroid pattern
Brim	5 mm when the part is tall or has a small footprint; otherwise off ABS: always on, 5 mm+ — it is the cheapest warp insurance there is
Seam	Aligned, with "wipe" on — hides the seam and kills blobs
Speed	Leave the profile's speeds alone — the volumetric cap above is the real limit

ABS changes three rules: shut the enclosure, turn the fan nearly off, and expect the part to come out slightly small. Everything that helps PLA — open door, full fan — is what cracks and warps ABS. Print it in a ventilated room: hot ABS gives off styrene.

Plate care: wash the textured PEI with dish soap & warm water weekly, wipe with IPA between, handle by the edges. Finger grease is the #1 reason a first layer won't stick.

New Spool & Troubleshooting

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EVERY NEW SPOOL — A 5-MINUTE ROUTINE

- 1 Load the spool. In Bambu Studio pick the **exact** filament profile (PLA Basic / PLA Matte / PETG / ABS) and set the **plate type** to Textured PEI.
For ABS: shut the door and top now — calibrate at the chamber temperature you'll print at.
- 2 Calibration tab → **Flow Dynamics** (pressure advance). Run it, accept the result.
- 3 Calibration tab → **Flow Rate**. Run it, apply the correction.
- 4 Slice a small test (a 20 mm cube or a real small part) and **watch the first layer** go down — it should look like smooth, touching lines with no gaps or ridges.
- 5 If the first layer looks off, nudge **Z-offset** in 0.02 mm steps (down = closer/flatter). Save the calibrated values to that filament so you only do this once per spool.

TROUBLESHOOTING QUICK-FIXES

Stringing / whiskers (usually PETG)

Dry the spool (65 °C / 8 h). Drop nozzle temp 5 °C. Confirm retraction is on and "wipe" is enabled.

Layers splitting / part snaps easily

Raise nozzle 5–10 °C. Lower the fan (PETG 20–30%). Use thinner layers (0.12–0.16 mm) for strength.

First layer won't stick

Re-clean the plate (soap + water). Add a 5 mm brim. Bed +5 °C. Slow the first layer to ~20 mm/s.

Corners lifting / warping

Add a brim. No cross-draft (don't point a fan at it). Keep the door shut for PETG. Bed +5 °C.

Elephant foot (squished first layers)

Enable elephant-foot compensation ~0.15 mm. Drop bed temp 5 °C. Check Z-offset isn't too low.

PLA Matte under-extrudes / clicks

Lower max volumetric speed to ~10 mm³/s. Make sure it's dry. Keep the enclosure open — it's heat-sensitive.

Blobs & zits on the surface

Seam = Aligned, "wipe" on, "seam not on overhangs" on. Run Flow Dynamics if you skipped it.

Ghosting / ripples after edges

Lower acceleration & speed in the profile. Make sure the printer sits on a solid, non-flexing surface.

ABS warps or peels off the plate

Door and top shut, and leave them shut — one door-open peek mid-print is enough to lift a corner. Brim 5 mm+. Bed 100 °C. Fan to 0% for the first 3 layers. Never aim a room fan at the printer.

ABS cracks between layers on tall parts

The chamber is too cool: shut it, wait for it to soak, print tall parts one at a time so each layer stays hot. Fan under 20%, nozzle +5–10 °C, 3 wall loops. Slower is warmer.

ABS part measures small / holes are tight

Normal — ABS shrinks ~0.5–0.8% as it cools. Scale the model up ~0.6%, or add 0.1–0.2 mm of clearance to sockets and pockets. Don't chase it with flow rate.

ABS smells strong / hazy chamber

Expected — that's styrene and ultrafine particles. Keep the enclosure closed, run the filter, ventilate the room, and don't sit beside it. This is a comfort *and* a health setting.

Golden rule: change **one** setting at a time and re-print the same small test. If you lose track, reset the filament profile to default and start from the preset again.